

FreshTrack

**Track your food.
Reduce waste.**

An AI-Powered Smart Food Inventory & Expiry Tracking
System

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Course: CSA-B | Mini-Project



The Global Food Waste Crisis

-  **Massive Scale:** Approximately 1.3 billion tons of edible food are discarded annually worldwide.
-  **Manual Tracking Fails:** Existing inventory apps require tedious manual entry that users cannot sustain long-term.
-  **Lack of Visibility:** Households discard substantial quantities of edible food simply because items are pushed to the back of the fridge and forgotten.
-  **The Solution Gap:** Users need automation that works seamlessly with their existing shopping behaviors, not against them.



The FreshTrack Solution



Automated OCR

Eliminate manual entry. Our Google ML Kit integration allows you to scan grocery receipts to automatically extract food item names and dates in seconds.



Proactive Alerts

Never forget expiring food again. The system calculates shelf life and sends timely push notifications before items spoil, driving behavioral change.



Cloud-Synced

Built with Supabase, your entire household inventory is securely synchronized to the cloud in real-time, persisting your data across all your devices.

Core Capabilities

Smart Date Suggestions

Automatically suggests accurate expiry dates based on a comprehensive, built-in shelf life dataset covering milk, eggs, chicken, fruits, and vegetables.

Expiry Status Tracking

Visual, color-coded badges make pantry audits instant. Know exactly what is Safe (🟢), Expiring Soon (🟡), and Expired (🔴) at a glance.

Post-Scan Review

After scanning a receipt, easily review a checklist of all extracted items to select or deselect exactly what you want to add to your digital pantry.

Search & Filter

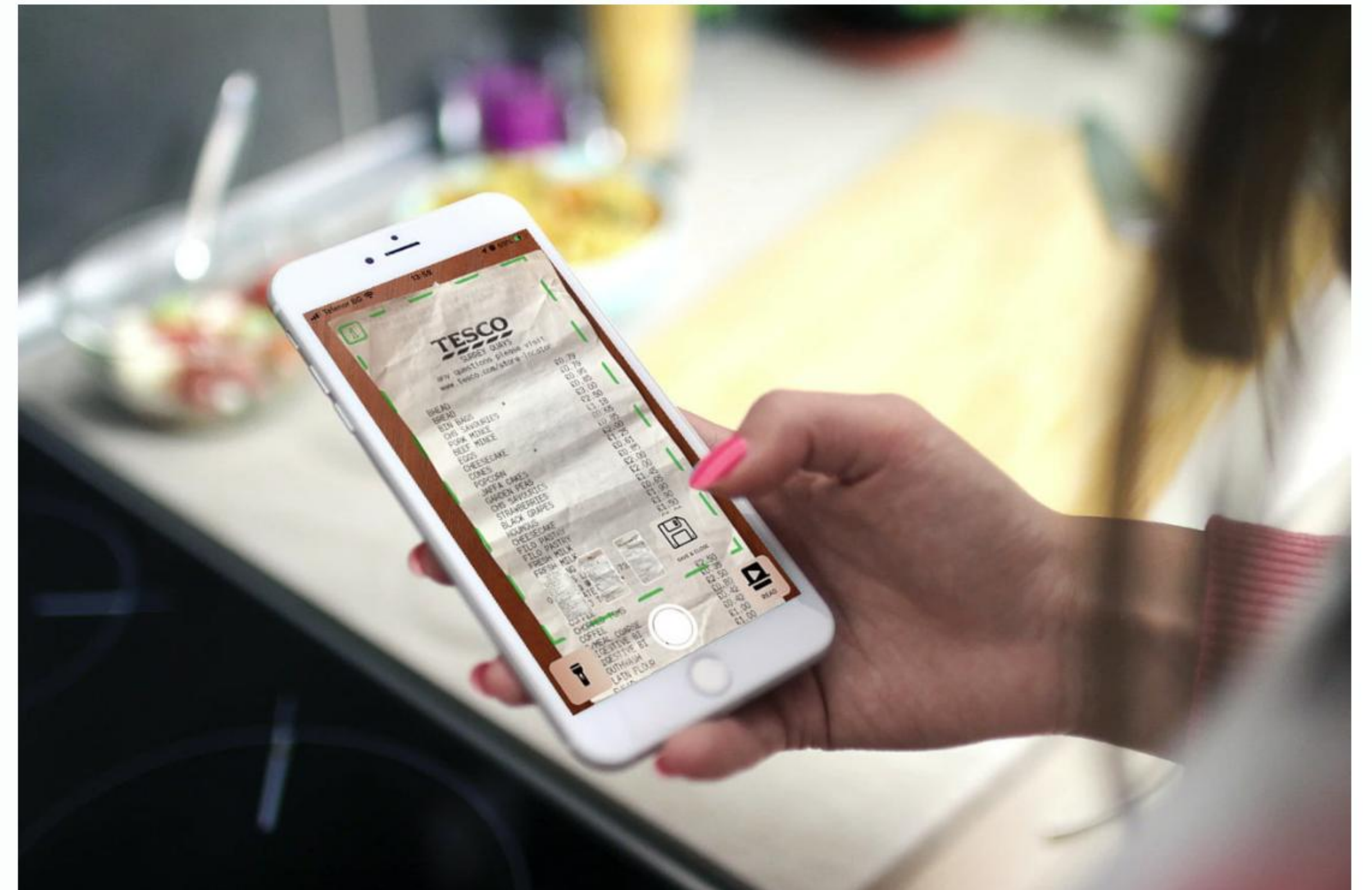
Quickly locate items using the search bar, or filter your inventory dynamically by "All", "Expiring Soon", or "Expired" to plan your next meal.

Modern Technology Stack

Layer	Technology	Purpose
Framework	Flutter (Dart)	Cross-platform mobile application development
Backend & Database	Supabase (PostgreSQL)	Robust cloud data storage and real-time synchronization
Authentication	Supabase Auth	Secure user identity management (Email/Password)
OCR / Machine Learning	Google ML Kit	On-device text recognition and receipt parsing
Notifications	Flutter Local Notifications	Scheduling exact alarms and timely proximity alerts

Intelligent Receipt Scanning

-  **Capture Image:** User snaps a photo of their grocery receipt using the device camera or gallery.
-  **ML Text Extraction:** Google ML Kit performs on-device Optical Character Recognition (OCR) to extract raw text blocks.
-  **Data Parsing:** Custom logic filters the text, separating store names and prices from actual food item names and dates.
-  **Review & Save:** Extracted items are presented in a checklist format for the user to confirm before saving to the cloud.



Smart Expiry Notifications

48Hrs

Advance Notice

Proactive Reminders

-  **Scheduled Alerts:** Users are automatically notified 2 days before an item expires, perfectly timed at 9:00 AM.
-  **Immediate Triggers:** Instant notifications are dispatched when adding or editing an item that already has ≤ 2 days left.
-  **Actionable Options:** Users can interact with the notification directly to select "Snooze" or mark the item as "I ate this".

Cloud-Synced Architecture

Seamless Data Flow

FreshTrack employs a modular architecture, keeping heavy data processing on-device while leveraging the cloud for persistence.



Client Layer: Flutter handles the UI, local state management, and notification scheduling.



AI Layer: ML Kit processes images locally, ensuring OCR data is parsed swiftly and securely.



Backend Layer: Supabase acts as the centralized PostgreSQL database, syncing inventory seamlessly across the user's devices.



Supabase Database Schema

Column Name	Data Type	Description
id	text (Primary Key)	Unique identifier for the food item record
name	text (Not Null)	The exact name of the scanned or entered food item
category	text (Not Null)	Classification group (e.g., Dairy, Fruits, Meat)
expiry_date	timestamptz (Not Null)	Timestamp recording the calculated date and time of expiration
quantity	text (Not Null)	String representing the amount (e.g., "1L", "500g")
user_id	uuid (Foreign Key)	References auth.users, handling data ownership per account

Security & Data Privacy



Robust Access Control

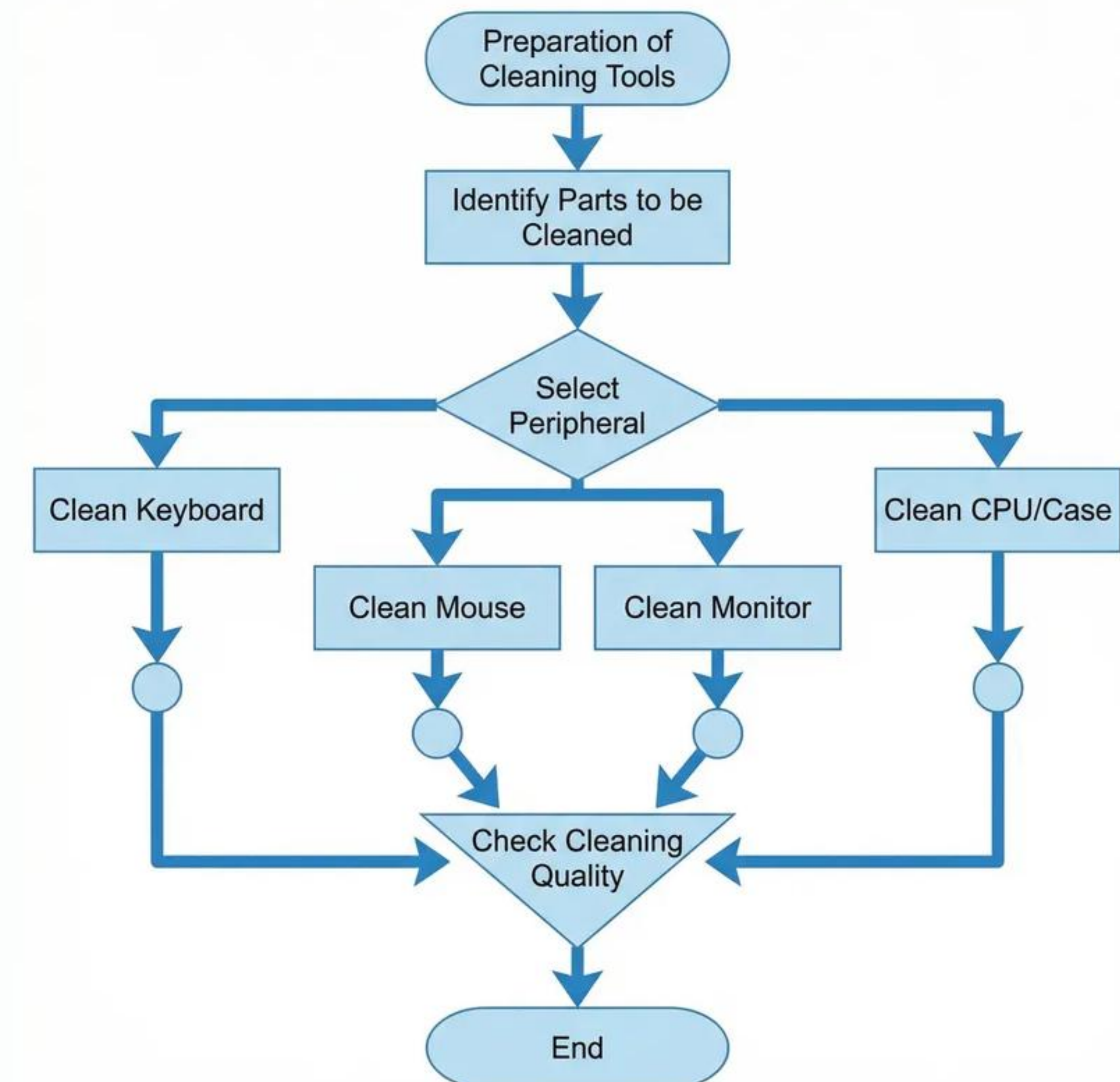
Security and data isolation are critical components of the FreshTrack infrastructure.

-  **Supabase Auth:** Implements secure email and password sign-in protocols, managing session tokens automatically.
-  **Row Level Security (RLS):** Database policies are strictly enforced at the PostgreSQL level. Users can only query, insert, or modify their own inventory rows based on their unique `auth.uid()`.

System Design Models

The development of FreshTrack is backed by comprehensive software engineering blueprints ensuring a scalable and maintainable codebase.

-  **Use Case & Activity:** Maps out the user journey from scanning receipts to saving items into the local database.
-  **Class & Sequence:** Details object orientation and the chronological interaction between the UI, OCR parser, and backend.
-  **Data Flow Diagrams:** DFD Levels 0, 1, and 2 break down the application's processes, mapping the exact flow of receipt data into structured database rows.



Conclusion & Future Outlook

Conclusion

FreshTrack successfully addresses the global food waste crisis at the household level. By combining cross-platform mobile technology (Flutter), machine learning (OCR), and robust cloud synchronization (Supabase), the application removes the friction of manual data entry and actively shifts user behavior through timely, automated expiry alerts.

Future Enhancements

-  Integrate Barcode Scanning for individual, non-receipt packaged goods.
-  AI Meal Planning to suggest recipes based specifically on items expiring soon.
-  Smart Refrigerator Integration via IoT sensors for fully automated tracking.

Thank You!

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github.com/Redanaz/FreshTrack_App

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